



Installation Instructions

Consolidated Energy Insulation

These instructions must be followed in conjunction with AS 3999:1992 or NZS 4246:2006, and all applicable local, state, and federal building regulations. Ensure compliance with site-specific safety requirements before commencing work.

Pre-Installation Requirements

- **Site Safety:** Confirm the roof space safety and assess the ceiling structure against building codes and project specifications.
- **Power Isolation:** Switch off mains power at the main switchboard. Consult a licensed electrician if uncertain.
- **Tools & Equipment:** Sharp knife, ladder, kneeling board, tape measure, non-conductive positioning stick and PPE.

LED downlights that can be safely covered with insulation must have an **IC-4** rating (sometimes marked as **IC** or **IC-F**). This rating allows them to be directly covered or abutted by insulation without posing a fire hazard.

Key Details Regarding IC-rated Downlights:

- **Safety Rating:** Only fixtures explicitly marked "IC-4" or "IC-F" are safe to cover, as they are designed to manage heat dissipation in restricted spaces.
- **Performance:** While safe, covering downlights may sometimes reduce bulb lifespan due to heat buildup.
- **Installation:** IC-rated lights allow for insulation to be laid continuously over the entire ceiling, improving energy efficiency.
- **Alternatives:** Non-IC-rated lights require a 50-100 mm clearance to prevent fire hazards.

Health & Safety Guidelines

- **Electrical Hazards:** Avoid contact with live wiring. Never use conductive fixing devices near electrical services. If in doubt, consult a licensed electrician.
- **PPE:** Wear long-sleeved clothing, gloves (AS2161), dust-resistant goggles (AS/NZS 1336), and a P2 dust mask.
- **Environmental Conditions:** Avoid installation during extreme heat and ensure adequate ventilation.
- Use correct lifting techniques at all times.

Installation Procedure

Step 1 – Product Selection

- Select insulation type and dimensions suitable for the ceiling framing.
- Where existing insulation is level with joists, install additional batts perpendicular to the original layer.
- Calculate material requirements: $\text{Area (m}^2\text{)} = \text{Length} \times \text{Width}$; $\text{Packs required} = \text{Total area} \div \text{m}^2 \text{ per pack}$.

Step 2 – Preparation

- Keep insulation in packaging until ready to install.
- Loosen batts by gently shaking to restore nominal thickness.
- Maintain clearance around downlights and heat sources per AS 3999/NZS 4246.

Step 3 – Installation

- Begin at the farthest corner, working toward the access point.
- Fit batts snugly between joists, ensuring there are no gaps.
- For truss roofs, butt batts firmly together to ensure continuous coverage.
- Maintain a 20 mm gap between insulation and roofing substrate at eaves.
- Cut batts carefully to avoid contact with electrical wiring.

Step 4 – Final Checks

- Verify complete and uniform coverage.
- Remove packaging and dispose of waste responsibly.

Step 5 – Access Panel

- Insulate the ceiling access hatch before exiting the roof space.
- Recommend secondary inspection to confirm compliance with applicable standards.

Post-Installation

- Reactivate power only after confirming safe conditions.
- If unable to restore power, seek assistance from a qualified electrician.